

Extending LLM Context Window with Adaptive Grouped Positional Encoding: A Training-Free Method

Xinhao Xu^{1,2*} Jiaxin Li^{1,2*} Hui Chen^{2†} Zijia Lin¹

Jungong Han^{2,3} Guiguang Ding^{1,2†}

¹School of Software, Tsinghua University, Beijing, China

²BNRist, Tsinghua University, Beijing, China

³Department of Automation, Tsinghua University, Beijing, China

xxhthu18@gmail.com, thulijx@gmail.com

jichenhui2012@gmail.com, dinggg@tsinghua.edu.cn

Abstract

Processing long input remains a significant challenge for large language models (LLMs) due to the scarcity of large-scale long-context training data and the high computational cost of training models for extended context windows. In this paper, we propose **Adaptive Grouped Positional Encoding** (AdaGroPE), a training-free, plug-and-play method to enhance long-context understanding in existing LLMs. AdaGroPE progressively increases the reuse count of relative positions as the distance grows and dynamically adapts the positional encoding mapping to sequence length, thereby fully exploiting the range of pre-trained position embeddings. Its design is consistent with the principles of rotary position embedding (RoPE) and aligns with human perception of relative distance, enabling robust performance in real-world settings with variable-length inputs. Extensive experiments across various benchmarks demonstrate that our AdaGroPE consistently achieves state-of-the-art performance, surpassing baseline methods and even outperforming LLMs inherently designed for long-context processing on certain tasks.

1 Introduction

Processing long input is essential for large language models (LLMs) (OpenAI, 2023b; Touvron et al., 2023a; Huang et al., 2025), enabling them to comprehend complex content such as academic papers, technical reports, and long-form dialogues, thereby expanding their applications in domains like healthcare, finance, and education (Wei et al., 2024; Lee et al., 2023; Xu et al., 2024; Shen et al., 2025). To support long-context processing, several LLMs with extended context windows have been developed (Chen et al., 2024b; Ruoss et al., 2023; Rozière et al., 2023). These models typically require fine-tuning with long sequences. Despite

exhibiting promising results, they rely on costly long-context dataset construction and require substantial GPU resources for training. Moreover, the scarcity of high-quality long-context data continues to limit their overall effectiveness (Gao et al., 2025).

To alleviate these constraints, recent training-free approaches have revealed that LLMs trained on short contexts can exhibit latent long-context processing capabilities (Xiao et al., 2024; Jin et al., 2024). For example, approaches such as StreamingLLM (Xiao et al., 2024) and LM-infinite (Han et al., 2024) manage the long-context challenge by restricting the number of neighbor tokens during inference to stay within the pre-trained attention window, improving perplexity on long-context tasks. However, these methods often discard significant context information and show limited effectiveness on real-world long-range dependent tasks. Other methods, such as SelfExtend (Jin et al., 2024) and An et al. (2024b), extend LLMs' context windows by reusing and remapping the position embedding from pre-training, achieving promising results on both language modeling and real-world long-context tasks without additional training.

In this paper, we propose a novel training-free framework, **Adaptive Grouped Positional Encoding** (AdaGroPE), to extrapolate LLM context windows. Building on the principles of rotary position embedding (RoPE) (Su et al., 2024), our approach dynamically adjusts position embedding according to sequence length and token distance, progressively increasing reuse count when calculating the position embedding of more distant tokens. This method draws inspiration from RoPE's long-term decay property, which allows the model to prioritize nearby tokens while paying less attention to those farther away. Furthermore, it aligns better with human perception of the long-context text, where the positions of nearby tokens are critical for

* Equal contribution

† Corresponding author

maintaining coherence and understanding, while distant tokens are processed more for their semantic content rather than their positions (Ivgi et al., 2023). To this end, AdaGroPE preserves fine-grained relative positions within a preset local window and introduces grouped reuse for farther tokens, such that the reuse count increases with distance. This design not only aligns with RoPE’s principle but also better reflects human’s processing of relative positions in long-context understanding, thereby enabling effective training-free extensions of LLM context windows.

AdaGroPE is a plug-and-play, training-free method that can be integrated into various LLMs. As a position embedding extension strategy, it complements and can be combined with other methods that enhance long-context understanding, such as fine-tuning on long-context datasets. Our method also stands out by dynamically adapting the position embedding mapping strategy to the input length, ensuring optimal performance in real-world scenarios with variable input lengths. We evaluate our approach across different LLMs and datasets, including language modeling, synthetic long-context tasks, and real-world long-context tasks. Experimental results show that AdaGroPE effectively extends the long-context understanding of LLMs with short-context windows, achieving state-of-the-art performance and even surpassing models natively designed for long-context processing. This demonstrates the potential of our approach to reduce reliance on expensive long-context training datasets.

In summary, our contributions are as follows:

1. We introduce a novel positional encoding strategy, AdaGroPE, which extends the range of pre-trained position embedding by gradually increasing the reuse count based on the tokens’ distance.
2. We implement AdaGroPE in a dynamic, adaptive adjusted manner, maximizing the use of pre-trained position embedding in real-world scenarios with variable input lengths.
3. We evaluate our method’s effectiveness across various long-context benchmarks and LLMs. Results show that AdaGroPE achieves state-of-the-art performance, even surpassing models with inherent long-context capabilities on certain tasks.

2 Method

2.1 Preliminary

In this section, we provide a brief overview of RoPE (Su et al., 2024), which serves as the foundation for our AdaGroPE method. RoPE is a crucial positional encoding mechanism designed to capture the relative positional relationships between tokens, which enhances the attention mechanisms in transformers. It extends traditional absolute position embeddings by incorporating positional information directly into the query and key vectors used in the self-attention process, allowing for more flexible handling of long sequences.

Let $\{\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{L-1}\}$ represent the token embeddings, where L is the sequence length, and each $\mathbf{x}_i \in \mathbb{R}^d$ is a d -dimensional vector. The key idea behind RoPE is to rotate the query $\mathbf{q}_i$ and key $\mathbf{k}_j$ vectors based on their positional indices i and j , so that the dot product $\mathbf{q}_i^T \mathbf{k}_j$ inherently captures the relative positional information between tokens. This is achieved by applying a complex rotation to the vectors $\mathbf{q}_i$ and $\mathbf{k}_j$. Specifically, for tokens at positions i and j , their corresponding query and key vectors are transformed as:

$$\mathbf{q}_i = f_q(\mathbf{x}_i, i), \mathbf{k}_j = f_k(\mathbf{x}_j, j), \quad (1)$$

where f_q and f_k represent the RoPE functions that apply the positional rotations. The resulting dot product between the query at position i and the key at position j depends solely on their relative positional difference $i - j$, ensuring that the model focuses on relative distances rather than absolute positions.

Specifically, RoPE constructs a relative position matrix M during self-attention, where each element $M[i][j] = i - j$ reflects the relative positional information between the i -th query and the j -th key. This matrix is structured as a Toeplitz matrix, where the same relative positions exhibit consistent values across rows and columns. Consequently, RoPE enables transformers to maintain strong relative position awareness without the need for explicit absolute position embeddings.

2.2 Progressive Reuse of Relative Positions with Increasing Count

We assume that the relative position matrix M has the i -th row denoted as m_i , with values ranging as follows for j from 0 to i :

$$m_i = [i, i - 1, \dots, 0], \quad (2)$$

where the maximum relative position is i . Suppose the largest observed context window during pre-training is w . When $i \geq w$, the inner product between $\mathbf{q}_i$ and $\mathbf{k}_j$ is computed using an out-of-distribution relative position encoding, which leads to performance degradation in LLMs.

To mitigate this, some methods have grouped relative positions by assigning the same relative position to neighboring tokens. These methods introduce a hyperparameter, the group size G_s , which controls the number of tokens in each group. As a result, the modified row m'_i becomes:

$$m'_i = [\lfloor \frac{i}{G_s} \rfloor, \lfloor \frac{i-1}{G_s} \rfloor, \dots, 0]. \quad (3)$$

By ensuring that $\lfloor \frac{i}{G_s} \rfloor < w$, out-of-distribution issues can be avoided, improving LLM performance on long-context tasks. Notably, the selection of G_s depends on both w and the length of the input.

As noted by SelfExtend (Jin et al., 2024), such a direct approach fails to account for the varying sensitivity between nearby tokens and distant tokens during contextual understanding. Specifically, it applies uniform reuse of relative positions regardless of token distance. To overcome this limitation, we propose a method that progressively increases the reuse count based on the relative distance between tokens.

In particular, given a target extension length L and a maximum relative position limit P , where $L > i$ and $P \leq w$, similar to SelfExtend, our approach defines a neighbor window size w_n . For tokens with relative distances smaller than the neighbor window size, which are more sensitive to positional information during contextual understanding, we retain their original relative positions:

$$M_a[i][j] = i - j \quad \text{if } i - j < w_n, \quad (4)$$

where M_a is the relative position matrix modified by our AdaGroPE method.

For tokens with relative distances greater than the neighbor window size, the values in M_a depend on L , P , and the hyperparameter reuse ratio coefficient $r = \frac{P}{w_n}$, following three guiding principles: *minimizing reuse*, *prioritizing distant relative position reuse*, and *progressively increasing reuse count from close to distant*. These principles will be illustrated in detail based on the progressive increase of L in the following sections. Figure 1 presents an example of the expansion of the relative position matrix $M_a[i][j]$ as L increases, with $P = 16$, $w_n = 4$, and $r = 0.25$.

Minimizing Reuse and Prioritizing Distant Relative Position Reuse

First, we define a sequence $\{L_n^{\max}\}_{n \in \mathbb{N}}$, representing the maximum allowable length L when the maximum reuse count required $G_s^m = n$. Naturally, $L_{n-1}^{\max} + 1$ denotes the minimum length L required for the maximum count $G_s^m = n$. Based on the definitions of $\{L_n^{\max}\}_{n \in \mathbb{N}}$ and w_n , we have:

$$\begin{aligned} L_1^{\max} &= P, \\ L_2^{\max} &= w_n + (P - w_n) \cdot 2 \\ &= 2P - w_n. \end{aligned} \quad (5)$$

This indicates that when $L_1^{\max} < L \leq L_2^{\max}$, the maximum reuse count required G_s^m is 2. Eq. (5) satisfies the principle of *minimizing reuse*, whereby relative position is reused only when L exceeds P , while simultaneously ensuring that the original relative position is employed for the nearest tokens, as shown in Eq. (4).

Furthermore, we define a sequence $\{d_n\}_{n \in \mathbb{N}}$, which represents the number of relative positions with a reuse count of $n + 1$ when $G_s^m = n + 1$:

$$d_n = L - L_n^{\max}. \quad (6)$$

It is evident that when $L_n^{\max} < L \leq L_{n+1}^{\max}$, at least d_n relative positions must be reused $n + 1$ times to ensure all relative positions remain within the maximum relative position limit P . In this context, when $L_1^{\max} < L \leq L_2^{\max}$,

$$M_a[i][j] = \begin{cases} i - j & \text{if } i - j < L - 2d_1, \\ f_1^m(i - j) & \text{others,} \end{cases} \quad (7)$$

where

$$f_n^m(x) = L - (n+1)d_n + \lfloor \frac{x - (L - (n+1)d_n)}{n+1} \rfloor. \quad (8)$$

$f_n^m(x)$ represents the mapping from the original to the AdaGroPE-adjusted relative position corresponding to the maximum relative position reuse count $G_s^m = n + 1$. This implies that the AdaGroPE-adjusted relative positions obtained through $f_n^m(x)$ all have a reuse count equal to the maximum required reuse count $n + 1$.

It is important to note that Eq. (8) groups the farthest positions and assigns the same relative position to $G_s^m = n + 1$ tokens within the group while preserving the relative positions of the $\mathbf{k}_j$ closer to $\mathbf{q}_i$. This *prioritization of distant relative position reuse* aligns with the notion that distant

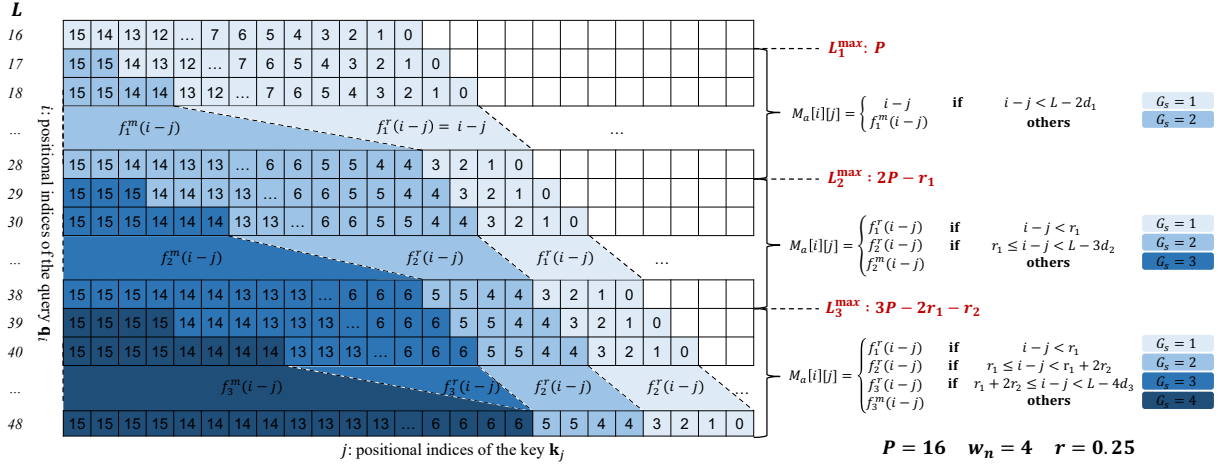


Figure 1: An example of the expansion of the computed relative positions in the AdaGroPE method. As L increases, the maximum reuse count of relative positions G_s^m increases from 1 to 4. The computed relative positions follow the principle of minimizing reuse, starting the reuse process from the farthest relative positions and progressing to the nearest. The reuse count increases with the distance from the current query.

tokens are less sensitive to positional encoding during attention calculation (Ivgi et al., 2023), and will continue to be reflected in the definition of M_a as L increases.

Progressive Increase in Reuse Count from Close to Distant The neighbor window size w_n preserves the original relative positions for keys k_j that are close to the query token q_i . As L increases, we progressively increase the reuse count from close to distant tokens.

Specifically, we define a sequence $\{r_n\}_{n \in \mathbb{N}}$ as follows:

$$r_n = \begin{cases} \lfloor \frac{r}{n} \cdot P \rfloor & \text{if } \log_2 n \in \mathbb{N}, \\ 0 & \text{otherwise,} \end{cases} \quad (9)$$

where r_n denotes the minimum number of relative positions retained with reuse count n when the maximum reuse count $G_s^m > n$ as L increases. The reuse ratio coefficient r is set to 0.25 by default. Eq. (9) ensures the number of relative positions retained for each reuse count G_s as L increases. These values decrease approximately geometrically as $\lfloor rP \rfloor$, $\lfloor \frac{rP}{2} \rfloor$, $\lfloor \frac{rP}{4} \rfloor$, etc., with only the reuse count corresponding to powers of 2 being retained.

Accordingly, we calculate $L_3^{\max}$:

$$\begin{aligned} L_3^{\max} &= r_1 + 2r_2 + (P - r_1 - r_2) \cdot 3 \\ &= 3P - 2r_1 - r_2. \end{aligned} \quad (10)$$

When $L_2^{\max} < L \leq L_3^{\max}$, the maximum reuse count G_s^m increases to 3. Consequently, the corresponding relative position matrix M_a is adjusted as

follows:

$$M_a[i][j] = \begin{cases} f_1^r(i-j) & \text{if } i-j < r_1, \\ f_2^r(i-j) & \text{if } r_1 \leq i-j < L-3d_2, \\ f_2^m(i-j) & \text{others,} \end{cases} \quad (11)$$

where

$$f_n^r(x) = \sum_{k=1}^{n-1} r_k \cdot k + \lfloor \frac{x - \sum_{k=1}^{n-1} r_n \cdot k}{n} \rfloor. \quad (12)$$

$f_n^r(x)$ inductively defines the mapping function from the original relative position to the AdaGroPE-adjusted relative position, corresponding to a reuse count less than or equal to n , under the condition that the maximum reuse count $G_s^m > n$. It is evident that $f_1^r(i-j) = i-j$ and the calculation formula for $M_a[i][j]$ in Eq. (7) when $i-j < L-2d_1$ also satisfies this function's definition.

Eq. (11) ensures that neighbor tokens around q_i retain their original relative positions, while reuse is progressively introduced for more distant tokens. The reuse count G_s increases in a controlled manner as defined in Eq. (9). Similarly to Eq. (7), Eq. (11) adheres to the principles of *minimizing reuse* and *prioritizing reuse for distant relative positions*. Besides, the relative positions retained by M_a follow the principle of *progressively increasing reuse count* as the distance from the q_i grows.

As L increases, AdaGroPE ensures that the adjusted relative positions follow the three guiding principles mentioned above. Similarly, we calculate $L_4^{\max}$ as follows:

$$\begin{aligned} L_4^{\max} &= r_1 + 2r_2 + 3r_3 + (P - r_1 - r_2 - r_3) \cdot 4 \\ &= 4P - 3r_1 - 2r_2 - r_3. \end{aligned} \quad (13)$$

To this end, when $L_3^{\max} < L \leq L_4^{\max}$,

$$M_a[i][j] = \begin{cases} f_1^r(i-j) & \text{if } i-j < r_1, \\ f_2^r(i-j) & \text{if } r_1 \leq i-j < r_1 + 2r_2, \\ f_3^r(i-j) & \text{if } r_1 + 2r_2 \leq i-j < L - 4d_3, \\ f_3^m(i-j) & \text{others.} \end{cases} \quad (14)$$

From Eq. (14), we deduce that when $L = L_4^{\max}$, reuse count of $G_s = 3$ are eliminated, leaving only $G_s = 1, 2, 4$, consistent with the rule in Eq. (9). Furthermore, when L increases to $L_4^{\max} + 1$, the maximum required reuse count G_s^m becomes 5, implying that the farthest relative position $P - 1$ is reused five times. We can iteratively obtain the subsequent adjusted relative positions by following the pattern established in Eq. (7), Eq. (11), and Eq. (14).

2.3 Adaptive Relative Position Adjustment Strategy

Based on the explanation above, extending L from values less than P up to $L_4^{\max}$, we have clarified the three fundamental principles of AdaGroPE’s relative position reuse, along with the intuitive process (illustrated in Figure 1). In this section, we will summarize the observed patterns and derive a direct formula for computing the relative position matrix using predefined values of L and P , demonstrating that our method can adaptively scale to longer target lengths L .

From Eq. (7), Eq. (11), and Eq. (14), we can derive the general expression for $L_n^{\max}$ as follows:

$$L_n^{\max} = nP - \sum_{k=1}^{n-1} (n-k-1)r_k. \quad (15)$$

Furthermore, based on the definition of $\{L_n^{\max}\}_{n \in \mathbb{N}}$, we obtain the formula for calculating the maximum reuse count G_s^m for varying lengths L as follows:

$$G_s^m(L) = \begin{cases} 1 & \text{if } L \leq L_1^{\max}, \\ 2 & \text{if } L_1^{\max} < L \leq L_2^{\max}, \\ \vdots & \vdots \\ n+1 & \text{if } L_n^{\max} < L \leq L_{n+1}^{\max}. \end{cases} \quad (16)$$

Finally, we derive the formula for calculating relative positions in AdaGroPE for any given target extension length L and a maximum relative

Notation	Explanation
i, j	Absolute position indices
$M_a[\cdot][\cdot]$	Relative position in AdaGroPE
L	Target context length after extension
w	Pre-trained context window length
P	Number of relative positions used; $P \leq w$
r	Coefficient controlling minimum retained positions per usage count
w_n	Size of neighbor window preserving original relative positions; $w_n = rP$
G_s	Usage count of relative positions; increases with position
$G_s^m(\cdot)$	Maximum usage count as a function of L with positions reused up to $G_s^m(L)$ times
$L_n^{\max}$	Maximum L for which $G_s^m(L) = n$, i.e., $G_s^m(L) = n$ holds iff $L \in (L_{n-1}^{\max}, L_n^{\max}]$
d_n	Number of positions used $n+1$ times when $G_s^m = n+1$; $L - L_n^{\max}$ for $L \in (L_n^{\max}, L_{n+1}^{\max}]$
r_n	Minimum retained positions used n times when $G_s^m > n$; $r_1 = w_n$

Table 1: Summary of notations and their corresponding explanations in AdaGroPE.

position limit P :

$$M_a[i][j] = \begin{cases} f_1^r(i-j) & \text{if } i-j < r_1, \\ f_2^r(i-j) & \text{if } r_1 \leq i-j < r_1 + 2r_2, \\ \vdots & \vdots \\ f_n^r(i-j) & \text{if } \sum_{k=1}^{n-1} kr_k \leq i-j < \sum_{k=1}^n kr_k, \\ \vdots & \vdots \\ f_{g_l-1}^r(i-j) & \text{if } \sum_{k=1}^{g_l-2} kr_k \leq i-j < \sum_{k=1}^{g_l-1} kr_k, \\ f_{g_l}^r(i-j) & \text{if } \sum_{k=1}^{g_l-1} kr_k \leq i-j < L - (g_l+1)d_{g_l}, \\ f_{g_l}^m(i-j) & \text{if } L - (g_l+1)d_{g_l} \leq i-j, \end{cases} \quad (17)$$

where $g_l = G_s^m(L) - 1$.

It is straightforward to verify that Eq. (7), Eq. (11), and Eq. (14) all satisfy the above equation. To this end, we finalize the construction of the AdaGroPE relative position matrix $M_a[i][j]$, which adheres to the three fundamental principles and can be directly calculated for any specified target extension length L and maximum relative position limit P . This provides a flexible and adaptive framework for configuring the positional encoding strategy. A detailed summary of the notations and their definitions, along with the pseudocode for computing relative positions during decoding, is provided in

Table 1 and Algorithm 1 in the Appendix, further clarifying the algorithmic process and implementation details of the proposed method.

3 Experimental Setup

Models and Baselines We evaluate our AdaGroPE on various LLMs: Llama-2 (7b and 13b) (Touvron et al., 2023b), Llama-3 (8b) (Dubey et al., 2024), Mistral (7b) (Jiang et al., 2023), SOLAR (10.7b) (Kim et al., 2024), and Phi-2 (Jawaheripati et al., 2023). In addition, we compare AdaGroPE’s performance with the other two state-of-the-art training-free long-context extension methods, Dual Chunk Attention (DCA) (An et al., 2024b) and SelfExtend (Jin et al., 2024). Furthermore, several models fine-tuned to extend their context windows, *i.e.*, LongLora (Chen et al., 2024b), Together (Together, 2023), CodeLlama (Rozière et al., 2023), and CLEX (Chen et al., 2024a) are included for comparison to demonstrate the superiority of AdaGroPE. All usages of scientific artifacts in this paper obey the corresponding licenses stated in the original papers or websites.

Datasets Following An et al. (2024b) and Jin et al. (2024), we present our main results on language modeling tasks, synthetic long-context tasks, and real-world long-context tasks. For language modeling, we use the PG19 (Rae et al., 2020) dataset, with context lengths ranging from 4k to 32k tokens. In synthetic long-context tasks, we include the passkey retrieval task, as defined in Landmark Attention (Mohtashami and Jaggi, 2023), where a language model must retrieve an n -digit passkey embedded within a long, meaningless text sequence. The passkey is placed at different depths within the document and tested across context lengths from 8k to 64k tokens. For real-world long-context tasks, we evaluate AdaGroPE on the LongBench (Bai et al., 2024) benchmark and four closed-ended tasks from L-Eval (An et al., 2024a): TOFEL, QuALITY (cleaned from Pang et al. (2022)), Coursera, and SFiction, following the setup of An et al. (2024b).

4 Main Results

Performance on Language Modeling Tasks We compute perplexity (PPL) for different models on the test data, where a lower PPL indicates better performance of LLMs. Table 2 shows that AdaGroPE achieves state-of-the-art performance across nearly

Model	Evaluation Context Window			
	4096	8192	16384	32768
Llama-2-7b	7.87	>100	>100	>100
ChunkLlama-2-7b	7.87	7.67	7.64	7.89
SE-Llama-2-7b	7.87	7.67	7.58	7.71
AdaGroPE-Llama-2-7b	7.87	7.65	7.56	7.75
LongLora-7b-32k*	8.14	7.85	7.70	7.80
Together-7b-32k*	8.21	7.95	7.76	7.64
CodeLlama-7b-16k*	8.93	8.64	8.44	8.36
CLEX-7b-16k*	16.74	15.08	14.28	14.70
Llama-3-8b	9.04	8.71	78.88	>100
Chunk-Llama-3-8b	9.04	8.71	8.61	8.62
SE-Llama-3-8b	9.13	8.80	8.59	8.52
AdaGroPE-Llama-3-8b	9.04	8.71	8.57	8.52

Table 2: Perplexity (PPL) ↓ evaluation on PG19 (Rae et al., 2020) validation set. We highlight the best results for each model size in bold. Models marked with * indicate those fine-tuned to extend their context windows.

all context lengths. Notably, we compare training-free methods with fine-tuned models designed to extend their context windows, marked with an asterisk (*). The results further demonstrate that the training-free AdaGroPE surpasses these training-dependent methods, underscoring the effectiveness of the proposed approach.

Performance on Synthetic Long Context Tasks

Figure 2 displays the evaluation results for various methods on the passkey benchmark (Mohtashami and Jaggi, 2023). In our experiments, the passkey consists of 36 digits, and we conduct multiple retrieval tests for each combination of context length and depth. The passkey is randomly placed within a 400-token span. For example, with a context length of 8k and a depth of 0.1, the passkey appears between tokens 800 and 1600. Each span is evaluated over 10 iterations, yielding 20 iterations in this setting.

As shown in Figure 2, AdaGroPE, without any fine-tuning, achieves nearly 100% passkey retrieval accuracy across all tested depths and context lengths. In comparison, the original Mistral-7b-instruct-0.1 with SWA sees a drastic performance drop to 0 at smaller depths, while ChunkMistral-7b-ins-0.1 displays significant accuracy fluctuations as the token limit increases. Although SelfExtend achieves results similar to AdaGroPE, its performance degrades at larger token limits, such as 65,536, where AdaGroPE consistently maintains superior accuracy.

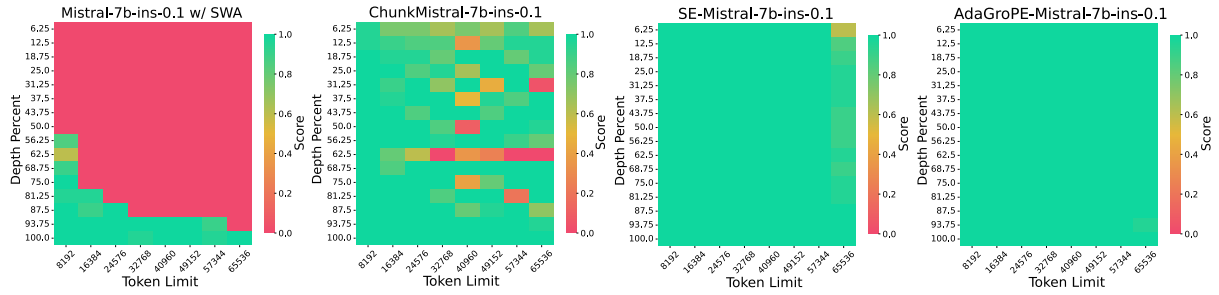


Figure 2: Passkey retrieval accuracy for Mistral-7b-instruct-0.1 with SWA, DCA, SelfExtend, or AdaGroPE. The number of passkey digits is set to 36. Mistral with AdaGroPE obtains nearly 100% passkey retrieval accuracy for all sequence lengths (token limits) and all depths.

Model	Tokens	Single-Document QA			Multi-Document QA			Summarization		Few-shot Learning			Synthetic		Code		Avg.	
		NarrativeQA	Qasper	MultiField-en	HopotQA	2WikiMQA	Musicqa	GovReport	QMSum	MultiNews	TREC	TriviaQA	SAMSum	PassageCount	PassageRe	Locc		
GPT-3.5-Turbo-16k	16k	23.6	43.3	52.3	51.6	37.7	26.9	29.5	23.4	26.7	68.0	91.4	41.7	4.5	71.0	54.7	53.6	43.7
ChatGLM2-6B-32k	32k	21.1	31.5	46.2	45.1	34.0	21.9	32.4	24.0	26.5	62.5	78.7	36.3	1.5	77.0	55.6	49.9	40.26
Baichuan-13B-4k	16k	0.07	17.55	17.28	3.29	15	0.1	6.8	1.71	23.1	20.05	20.06	5.77	0.06	0.5	47.98	16.58	10.49
ALiBi-7B-4k	16k	0.04	8.13	17.87	2.73	8	1.33	5.31	1.64	25.55	9.25	8.83	4.67	0	1.27	46.69	18.54	9.48
Llama-2-7b-chat	4k	18.7	19.2	36.8	25.4	32.8	9.4	27.3	20.8	25.8	61.5	77.8	40.7	2.1	9.8	52.4	43.8	31.52
ChunkLlama-2-7b-chat	25k	20.27	25.80	34.87	23.18	28.42	10.15	27.03	21.27	26.47	68.50	74.86	41.51	1.48	4.75	58.05	50.76	32.34
SE-Llama-2-7b-chat	25k	21.37	26.68	34.63	35.47	30.46	15.51	27.51	21.30	25.87	68.50	78.79	41.29	3.90	3.50	59.69	53.83	34.26
AdaGroPE-Llama-2-7b-chat	25k	18.72	29.61	40.20	37.33	30.86	15.73	28.39	21.45	26.10	69.50	83.23	42.16	3.48	6.00	59.38	52.24	35.27
Llama-3-8b-ins	8k	21.63	44.11	44.35	46.84	35.84	21.53	29.98	22.66	27.75	75.50	90.58	42.67	6.50	66.50	56.81	51.24	42.78
ChunkLlama-3-8b-ins	25k	26.54	42.36	47.82	47.54	35.27	25.19	31.86	23.02	27.39	77.00	90.31	42.81	7.00	75.50	58.49	55.15	44.58
SE-Llama-3-8b-ins	25k	23.88	43.82	50.64	50.71	36.58	30.24	32.90	23.90	27.79	76.00	91.68	42.93	4.40	98.00	56.72	47.53	46.11
AdaGroPE-Llama-3-8b-ins	25k	25.72	44.09	51.80	52.10	38.80	31.82	32.81	24.11	27.90	76.50	91.13	42.30	7.62	99.00	56.49	51.46	47.10
Mistral-7b-ins-0.1 w/o SWA	8k	20.46	35.36	39.39	34.81	29.91	11.21	24.70	21.67	26.67	68.00	86.66	41.28	0.18	24.00	56.94	55.85	36.07
Mistral-7b-ins-0.1 w/ SWA	16k	19.40	34.53	37.06	42.29	32.49	14.87	27.38	22.75	26.82	65.00	87.77	42.34	1.41	28.50	57.28	53.44	37.08
ChunkMistral-7b-ins-0.1	16k	20.86	36.56	42.40	35.89	31.25	12.47	28.08	22.87	27.09	69.50	86.52	42.94	2.14	21.50	54.92	52.70	36.73
SE-Mistral-7b-ins-0.1	16k	23.56	39.33	49.50	45.28	34.92	23.14	30.71	24.87	26.83	69.50	86.47	44.28	1.18	29.50	55.32	53.44	39.86
AdaGroPE-Mistral-7b-ins-0.1	16k	25.02	39.00	53.38	47.88	35.26	25.47	31.26	23.84	26.67	70.50	86.66	43.86	3.41	33.50	55.05	51.50	40.77
SOLAR-10.7b-ins	4k	16.50	24.06	46.76	44.03	36.05	22.76	31.39	19.81	26.36	70.00	87.91	42.49	4.50	26.50	41.04	54.36	37.16
ChunkSOLAR-10.7b-ins	16k	22.48	29.77	48.84	51.62	34.80	27.35	31.59	21.75	26.22	74.50	87.41	42.69	7.50	20.00	48.98	54.94	39.40
SE-SOLAR-10.7b-ins	16k	22.63	32.49	47.88	46.19	34.32	27.88	30.75	22.10	25.62	74.50	89.04	42.79	4.00	28.00	53.73	56.47	39.40
AdaGroPE-SOLAR-10.7b-ins	16k	24.35	34.42	48.81	53.31	43.30	33.93	32.38	22.29	26.51	74.50	89.62	43.11	6.50	36.00	54.32	58.64	42.62
Phi-2	2k	4.46	7.01	19.98	9.43	8.55	4.62	25.64	14.32	24.03	50.50	74.55	1.71	2.83	4.17	58.96	54.14	22.81
SE-Phi-2	8k	12.04	12.10	20.15	8.22	9.68	3.89	27.90	14.58	22.13	61.00	82.82	1.40	2.37	2.83	57.87	56.42	24.71
AdaGroPE-Phi-2	8k	14.14	11.90	26.80	9.96	11.37	5.09	29.68	20.04	25.19	60.00	82.69	1.29	2.37	4.73	58.10	55.07	26.15

Table 3: Performance comparison of different LLMs on LongBench (Bai et al., 2024). Best and second-best results in each group are highlighted with bold and underline, respectively. The same applies below.

Performance on Real-World Long Context Tasks

The evaluation results on LongBench and L-Eval are shown in Table 3 and Table 4, respectively. Following Jin et al. (2024), we present the performance of representative large language models (OpenAI, 2023a; Zeng et al., 2024; Baichuan, 2023) for reference, including those employing the ALiBi position encoding scheme (Press et al., 2022), on the LongBench benchmark. These results are reported by the LongBench (Bai et al., 2024) and CLEX (Chen et al., 2024a). As illustrated in Table 3, AdaGroPE significantly enhances the performance of the original models and outperforms other training-free extension methods, such as SelfExtend (Jin et al., 2024) and DCA (An et al., 2024b), achieving the best overall average performance. Table 4 further highlights AdaGroPE’s superior performance and broad applicability. Notably, it demonstrates that Ada-

GroPE enables models with smaller initial context windows to exceed the performance of models with inherently larger context windows, which are pre-trained or fine-tuned for long-text understanding. For instance, AdaGroPE-Llama-2-7b-chat and AdaGroPE-Vicuna-2-1.5-7b, both based on models with 4k context windows, achieve better average performance than Longchat-1.5-7b-32k and Vicuna-1.5-7b-16k, respectively. Without relying on fine-tuning or additional training, AdaGroPE achieves competitive performance during inference, highlighting its potential as a resource-efficient approach for extending the context windows of existing LLMs.

5 Analysis

Performance as the Context Length Increases

Figure 3 presents the performance of three different training-free methods for extending long texts

Model	TOFEL QuALITY Coursera SFiction				Avg.
	(3k~5k)	(4k~9k)	(5k~17k)	(6k~27k)	
Llama-2-7b-chat	51.67	37.62	29.21	60.15	44.66
Longchat-1.5-7b-32k	39.77	37.62	32.99	57.02	41.85
ChunkLlama-2-7b-chat	<u>57.62</u>	35.14	32.12	<u>61.72</u>	46.65
SE-Llama-2-7b-chat	55.39	41.09	35.76	57.81	<u>47.51</u>
AdaGroPE-Llama-2-7b-chat	61.34	<u>38.12</u>	<u>35.47</u>	64.06	49.28
Llama-2-13b-chat	60.96	<u>42.57</u>	35.75	54.68	48.49
ChunkLlama-2-13b-chat	<u>66.54</u>	43.06	<u>41.56</u>	57.03	52.05
SE-Llama-2-13b-chat	66.17	41.09	38.95	63.28	<u>52.37</u>
AdaGroPE-Llama-2-13b-chat	68.77	40.59	46.66	<u>57.81</u>	53.46
Vicuna-1.5-7b-16k	<u>55.39</u>	39.60	<u>38.66</u>	60.15	48.45
SE-Vicuna-1.5-7b	<u>55.39</u>	41.58	37.21	63.28	<u>49.37</u>
AdaGroPE-Vicuna-1.5-7b	56.51	41.58	42.01	<u>60.94</u>	50.26
SOLAR-10.7b-ins	77.32	59.90	48.84	69.53	63.90
SE-SOLAR-10.7b-ins	<u>79.18</u>	70.30	<u>50.44</u>	73.44	<u>68.34</u>
AdaGroPE-SOLAR-10.7b-ins	81.78	<u>68.81</u>	56.83	<u>71.88</u>	69.83
Phi-2	55.76	42.08	38.37	<u>52.34</u>	47.14
SE-Phi-2	62.83	41.08	42.44	<u>52.34</u>	<u>49.67</u>
AdaGroPE-Phi-2	68.40	<u>41.58</u>	<u>41.28</u>	55.47	51.68

Table 4: Comparison with open-source chat models and proprietary models on 4 closed-ended tasks with various input lengths from L-Eval (An et al., 2024a).

as the input length varies, specifically for passkey lengths of 16 digits, 48 digits, and 64 digits. Notably, AdaGroPE maintains an accuracy of over 90% as the context window lengthens, in contrast to ChunkMistral and SE-Mistral, whose performance exhibits significant degradation with increasing context window sizes. This degradation is particularly pronounced when the passkey is set to 64 digits, where the accuracy of ChunkMistral and SE-Mistral declines from approximately 100% and 80% at an 8k context window to around 60%. The comparison results suggest that AdaGroPE demonstrates more robust performance when handling longer context windows, especially in more challenging tasks.

Ablation Studies on the Selection of P and r
Figure 4 presents the results of the ablation study on the selection of P and r . Following An et al. (2024b), we conduct experiments on two real-world datasets from Longbench: Qasper and Lcc.

As shown in Figure 4, the selection of P has a more significant impact on the performance of AdaGroPE, with the optimal P varying across different tasks. This reflects a trade-off between leveraging more comprehensive positional encodings and the degree of pre-training on these encodings. On one hand, larger P values allow for better utilization of the relative positional encodings learned during

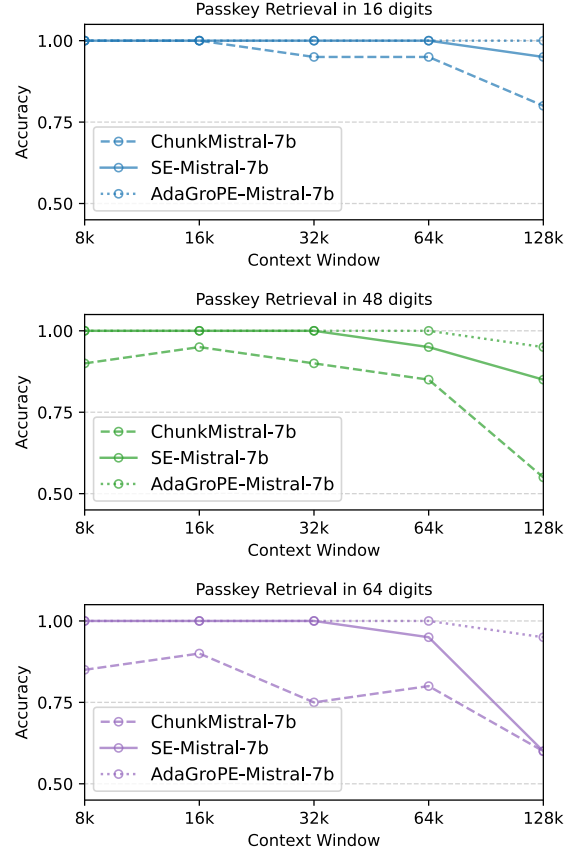


Figure 3: The performance of different training-free context window extension methods as the context length increases. AdaGroPE demonstrates robust performance in passkey retrieval as the input length increases, particularly when the number of digits in the passkey increases.

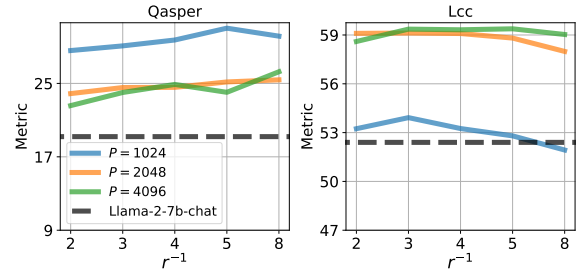


Figure 4: Ablation study results on the selection of P and r . The selection of P has a more significant impact on the method’s performance compared to that of r .

pre-training, resulting in more accurate relative position representation. On the other hand, as the relative position range extends, the lack of sufficient pre-training for larger positions can lead to degradation in performance. Notably, Lcc, a code-based dataset, is sensitive to the relative positions of nearby tokens and tends to benefit from more precise encodings. This may explain why larger

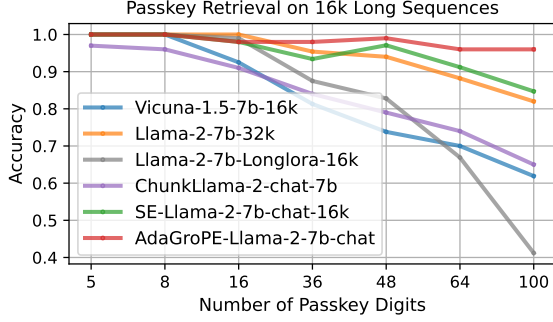


Figure 5: Passkey retrieval accuracy for fine-tuned long-context models and training-free context extension methods on Llama-2-chat-7b.

values of P and w_n generally lead to improved performance, as reflected by the green and orange lines outperforming the blue line in the right figure. Additionally, we observe that the optimal r varies across datasets. We find that our default setting $r = 0.25$ consistently yields strong average performance and provides reliable improvements over the Llama-2-7b-chat baseline.

Varying-Length Passkey Retrieval Task We validate AdaGroPE’s capability to capture information in long contexts by conducting experiments on the passkey retrieval task, with passkey digit lengths set to 5, 8, 16, 36, 48, 64, and 100, respectively. As the number of digits increases, the task complexity also increases accordingly.

As shown in Figure 5, the performance of ChunkLlama, Longlora, and Vicuna declines significantly as the number of passkey digits increases, especially beyond 8 digits. Despite Vicuna and Longlora being fine-tuned for long-context windows, they still struggle with more difficult passkey retrieval tasks that demand higher precision. In contrast, while all methods exhibit some performance degradation, AdaGroPE shows a notably milder decline and maintains relatively robust overall results. These findings suggest the potential of AdaGroPE as an effective training-free alternative for long-context modeling, while also highlighting the challenges that fine-tuning-based methods may face in accurately capturing information across extended sequences. The performance of the varying-length passkey retrieval task on longer sequences is provided in the Appendix.

6 Conclusion

In this paper, we propose a novel long-context window extension method, AdaGroPE, which can be applied to existing LLMs with short-context windows in a training-free, plug-and-play manner. AdaGroPE employs a progressively reused relative position encoding strategy, adhering to three key principles when constructing the relative position matrix: minimizing reuse, prioritizing reuse of distant relative positions, and progressively increasing reuse count from nearby to distant positions. This adaptive approach allows the relative position matrix to be tailored to the target context window length. We demonstrate the effectiveness and superiority of AdaGroPE across language modeling tasks, synthetic long-context tasks, and real-world long-context tasks, and further validate its robustness under increasing context lengths and task complexity.

Limitations

The proposed AdaGroPE is an empirically validated method for improving long-context processing in large models. While its effectiveness has been demonstrated through extensive experiments, a more thorough theoretical analysis of the underlying principles behind positional encoding in large language model attention mechanisms is not included. We believe future work should delve deeper into the intrinsic mechanisms of transformer positional encodings to develop novel approaches for enhancing long-text understanding in large language models. Additionally, our current exploration focuses solely on extending long-sequence capabilities in single-modal settings. Further investigation and validation are needed for multimodal approaches that integrate modalities such as images, videos, and audio.

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A Related Work

A.1 Positional Encoding

Position information is crucial for transformer models (Vaswani et al., 2017; Xiong et al., 2025; Wang et al., 2024; Xu et al., 2025) and is commonly represented using either absolute or relative embeddings. Absolute position embeddings assign a vector based on a token’s position within the sequence, as seen in both sinusoidal and learned embeddings, such as those used in GPT-3 (Brown et al., 2020) and OPT (Zhang et al., 2022). In contrast, relative positional encodings, which have become widely adopted, capture token distances relative to one another, enhancing contextual understanding, particularly in long-context scenarios. Notable approaches include RoPE (Su et al., 2024) and ALiBi (Press et al., 2022), which have been incorporated into prominent models like Llama (Touvron et al., 2023a) and Falcon (Penedo et al., 2023). Our method builds upon the RoPE framework, aiming to optimize positional encoding for more effective long-context modeling.

A.2 Extrapolation of RoPE

Research has shown that directly extrapolating RoPE leads to significant performance degradation in long-context tasks (Chen et al., 2023; Chowdhury and Caragea, 2023; Chen et al., 2024a), primarily due to the model encountering unseen relative positions during pre-training (Jin et al., 2024). To address this, recent approaches have focused on training techniques that enhance the long-context understanding of LLMs after extrapolation (Rozière et al., 2023; Together, 2023; Lian et al., 2025). In addition, some studies explore training-free methods that ensure relative positions remain within the scope of the observed context length (An et al., 2024b; Jin et al., 2024; Chen et al., 2023; LocalLLaMA, 2023), thus reusing position embeddings and mitigating extrapolation-related degradation. However, these methods often fail to account for variations in relative positions and struggle to adapt dynamically to changing input lengths. In contrast, AdaGroPE builds on these prior techniques, offering a training-free, plug-and-play solution that distinguishes itself through the progressive reuse of relative positions and its dynamic adaptability to input length variations.

Model	P	TOFEL (3k~5k)	QuALITY (4k~9k)	Coursera (5k~17k)	SFiction (6k~27k)
Llama-2-7b-chat	N/A	51.67	37.62	29.21	60.15
	1k	61.34	36.63	37.50	60.15
+AdaGroPE	2k	56.13	38.12	35.47	64.06
	4k	53.53	37.62	36.05	61.03

Table 5: Impact of P on the effectiveness of AdaGroPE across tasks with different context lengths.

B Implementation Details

All the experiments in the paper are conducted on a single NVIDIA H800 GPU. Unless otherwise specified, all experimental results in the paper are based on the default setting of $r = 0.25$. We find that this default setting is generally applicable across different tasks. Furthermore, as noted in Section 2.2, w_n can be parameterized by P and r , with w_n set to $0.25P$ in all experiments by default.

For the selection of P , we set $P = w$ on language modeling tasks, while for the other tasks, we generally set $P = w/2$, where w is the pre-training context window size of the model. As highlighted in prior work (Jin et al., 2024), positional encodings with smaller relative distances are more effectively trained. Therefore, extrapolation based on smaller relative positions tends to yield better performance. Besides, for tasks where the input context length is relatively short (*e.g.*, close to or less than the pre-training window size), we observe that limiting P to a smaller range, such as $w/4$, leads to better results. Taking the experiments on L-Eval in Table 4 as an example, we set $P = w/4$ for the TOEFL (3k–5k) task and $P = w/2$ for the other three tasks: QuALITY (4k–9k), Coursera (5k–17k), and SFiction (6k–27k). Table 5 illustrates the effects of applying AdaGroPE with different P values on Llama-2-7b-chat (pre-training window size $w = 4k$).

We observe that the proposed guideline for setting P achieves optimal average performance and we report only the results obtained using the P values selected according to this guideline. Although a more refined selection of P for specific tasks may yield better results in some cases (*e.g.*, in the Coursera task, AdaGroPE performs better when $P = w/4 = 1k$), we believe that following the proposed guideline ensures the effectiveness of AdaGroPE.

Algorithm 1 provides the pseudocode for relative position computation in AdaGroPE during decod-

Algorithm 1 Pseudocode for relative position computation in AdaGroPE during decoding

Input: L $\triangleright$ Target context length after extension
 P $\triangleright$ Number of relative positions used
 r $\triangleright$ Reuse ratio coefficient

Output: m_a $\triangleright$ Relative position in AdaGroPE

```

1: function ADAGROPERELPOS( $L, P, r$ )
2:   if  $L \leq P$  then
3:      $m_a \leftarrow [L-1, L-2, \dots, 0]$ 
4:     return  $m_a$ 
5:   end if
6:    $G_s^m \leftarrow 1$ 
7:    $L_n^{\max} \leftarrow P$ 
8:    $r_{\text{sum}}, l_{\text{sum}} \leftarrow 0$ 
9:    $m_a \leftarrow []$ 
10:  while  $L_n^{\max} < L$  do
11:    if isPowerOfTwo( $G_s^m$ ) then
12:       $r_n \leftarrow \lfloor r \times P / G_s^m \rfloor$ 
13:      for  $p = r_{\text{sum}}$  to  $r_{\text{sum}} + r_n - 1$  do
14:        prepend  $p$  to  $m_a$  for  $G_s^m$  times
15:      end for
16:       $r_{\text{sum}} \leftarrow r_{\text{sum}} + r_n$ 
17:       $l_{\text{sum}} \leftarrow l_{\text{sum}} + r_n \times G_s^m$ 
18:    end if
19:     $G_s^m \leftarrow G_s^m + 1$ 
20:     $L_n^{\max} \leftarrow (P - r_{\text{sum}}) \times G_s^m + l_{\text{sum}}$ 
21:  end while
22:   $d_{n-1} \leftarrow P - (L_n^{\max} - L) - r_{\text{sum}}$ 
23:  for  $p = r_{\text{sum}}$  to  $P - d_{n-1} - 1$  do
24:    prepend  $p$  to  $m_a$  for  $G_s^m - 1$  times
25:  end for
26:  for  $p = P - d_{n-1}$  to  $P - 1$  do
27:    prepend  $p$  to  $m_a$  for  $G_s^m$  times
28:  end for
29:  return  $m_a$ 
30: end function

```

ing. Here, m_a denotes the relative position with respect to the query token, while l_{sum} and r_{sum} denote, respectively, the total number of true relative positions that have been mapped to computed relative positions in AdaGroPE, and the total number of relative positions used that correspond to these mapped true relative positions. Specifically, r_{sum} corresponds to the summation of the terms defined by Eq. (9). The implementation proceeds by first iteratively determining the maximum reuse count G_s^m , during which relative positions with reuse counts less than G_s^m are retained according to Eq. (9). Finally, based on the target extension length L , the most distant relative positions reused

Model	Latency (s/token)			
	32k	64k	96k	128k
Llama-2-7b-chat	1.81	5.30	10.62	17.77
AdaGroPE-Llama-2-7b-chat	2.10	5.80	11.24	18.48

Model	Memory (MB)			
	32k	64k	96k	128k
Llama-2-7b-chat	28787	43925	59063	74193
AdaGroPE-Llama-2-7b-chat	29543	45433	61323	77201

Table 6: Comparison of latency and memory consumption across context lengths between the original model and our AdaGroPE.

$G_s^m - 1$ and G_s^m times are added to the resulting relative position sequence.

C Latency and Memory Analysis

We also conduct comparisons on the passkey retrieval task across different context length settings, evaluating the token generation latency (s/token) and GPU memory consumption (MB) for the original model and our proposed method. The results are shown in Table 6.

We observe that AdaGroPE increases inference latency and memory usage by no more than 10% on average compared to the original model. This demonstrates the practical applicability of the proposed method in real-world scenarios.

D Limitations in Challenging Tasks

Despite the overall effectiveness of our method, we observe certain failure cases that reveal its current limitations. As shown in Table 3, AdaGroPE performs suboptimally on code tasks on average. We believe this is due to the distinct characteristics of code compared to natural language. Specifically, the assumption that distant relative positions can be less precise than closer ones, which generally holds for natural language, does not strictly apply to code. Code understanding requires a higher degree of accuracy in relative distances between tokens, as it is significantly influenced by the structural semantic relationships and hierarchical organization between tokens. These factors cannot be inferred solely from relative distance.

Beyond this, by examining the outputs of QA tasks in the LongBench benchmark, we observe that AdaGroPE’s performance tends to decline on questions requiring reasoning, inference, or complex information integration across long contexts.

Model	Position Emb	Training context	Context Window			
			32k	64k	96k	128k
CodeLlama 7b	NTK	16k	8.36	8.65	9.14	9.87
+AdaGroPE	NTK	16k	8.34	8.32	8.38	8.48
Together 7b	PI	32k	7.64	>100	>100	>100
+AdaGroPE	PI	32k	7.64	7.58	7.58	7.60

Table 7: Integration of AdaGroPE with other long context window extension methods.

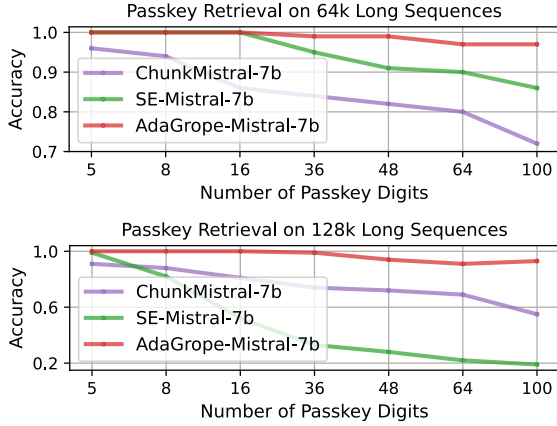


Figure 6: Passkey retrieval accuracy with longer sequence lengths on Mistral-7b.

Specifically, we find that the model’s responses to "Why" questions are often less satisfactory compared to "What" or "Where" questions, which are easier to locate answers directly. Although AdaGroPE enables the model to handle and understand contexts beyond the pre-training window, it seems to not fully resolve the original model’s limitation in precise reasoning and analysis of long-distance information.

These failure cases reveal opportunities for improving the method, such as enabling finer-grained control over position reuse or integrating auxiliary mechanisms to better handle extremely long contexts. We leave these directions for future investigation.

E Additional Experimental Results

E.1 Performance of Integrating AdaGroPE with Other Long-Context Window Extension Methods

Table 7 presents the performance of AdaGroPE applied to Codellama (Rozière et al., 2023) and Together (Together, 2023). Codellama and Together expand their context windows to 16k and 32k, re-

Model	MultiField-en	2WikiMQA	GovReport
SE-Phi-2	26.33	11.33	27.99
AdaGroPE-Phi-2	27.79	12.49	29.68

Model	TrivialQA	PassageCount	RepoBench-P
SE-Phi-2	83.14	2.12	52.82
AdaGroPE-Phi-2	84.71	3.09	53.63

Table 8: Evaluation results on LongBench (Bai et al., 2024) conducted on Ascend 910 GPUs.

spectively, using NTK (LocalLLaMA, 2023) and PI (Chen et al., 2023) strategies. The table shows the PPL results (Rae et al., 2020), demonstrating that AdaGroPE can effectively integrate with existing long-context expansion methods, further enhancing the language modeling capabilities of models with already large context windows.

E.2 Performance of AdaGroPE on Varying-Length Passkey Retrieval Task with Longer Sequence Lengths

To further evaluate the performance of AdaGroPE and baseline methods, we extended the input sequence length to 64k and 128k. As shown in Figure 6, AdaGroPE exhibits a noticeably slower degradation with increasing numbers of passkey digits at the longer 64k and 128k input sequence lengths, outperforming baseline methods. This highlights the dual advantages of the proposed method in handling both longer input sequences and larger numbers of passkey digits.

E.3 Evaluation on Ascend 910 GPUs

We validate the effectiveness of the AdaGroPE on Ascend 910 GPUs, as presented in Table 8. AdaGroPE outperforms the baseline on datasets spanning diverse sub-tasks.